

Genetic Algorithm — Typed Notes

Typed-text version prepared from the uploaded source PDF. Original terminology and structure are retained as closely as possible.

Genetic Algorithm

Search Methods

- Deterministic → Numerical Methods; Graph Search Methods.
- Graph Search Methods → Blind Search: Depth-first, Breadth-first, Dijkstra.
- Graph Search Methods → Informed Search: Hill climbing, Beam search, Best-first, A*.
- Stochastic → Trajectory Methods: TS, SA.
- Stochastic → Population-based Methods → Evolutionary Computing: GA; Swarm Intelligence: PSO, ACO.

Genetic Algorithm — Background

- Developed by John Holland, University of Michigan (1970's).
- To understand the adaptive processes of natural systems.
- To design artificial systems software that retains the robustness of natural systems.
- The original form of the GA, as illustrated by John Holland in 1975, had distinct features: a bit string representation, proportional selection and cross-over to produce new individuals.
- Several variations to the original GA have been developed using different representation schemes, selection and reproduction operators.

Genetic Algorithm — Basic Idea

- A class of probabilistic optimization algorithms.
- Inspired by the biological evolution process.
- Uses concepts of “Natural Selection” and “Genetic Inheritance” (Darwin 1859).
- Take a population of candidate solutions to a given problem.
- Use operators inspired by the mechanisms of natural genetic variation.
- Apply selective pressure toward certain properties.
- Evolve a more fit solution.

The Algorithm

Diagram / figure text: Initialization → Population → Parent selection → Parents → Cross-over + Mutation → Offspring → Survivor selection → Population; repeat by generation until termination.

Steps in the Genetic Algorithm

1. Initialization

The algorithm starts with a random population of possible solutions (individuals).

2. Parent Selection

The best individuals (parents) are selected based on their fitness (how good they are at solving the problem).

3. Crossover & Mutation

Crossover: The selected parents combine their traits (genes) to create new solutions (offspring).

Mutation: Some small random changes are introduced to increase diversity and avoid getting stuck in bad solutions.

4. Offspring Generation

The newly created offspring replace weaker individuals in the population.

5. Survivor Selection

The algorithm chooses the best individuals from both parents and offspring to form the next generation.

6. Iteration (Generation Loop)

Steps 2–5 repeat for multiple generations until the best solution is found.

7. Termination

The process stops when the algorithm meets a stopping condition, such as:

- The best solution doesn't improve after many generations.
- A fixed number of generations is reached.
- The solution reaches a desired quality level.

Genetic Algorithm — Evolutionary View

- Solution to a problem solved by genetic algorithms is evolved.
- Algorithm is started with a set of solutions (represented by chromosomes) called Population.
- Solutions from one population are taken and used to form a new population for a better one.
- Solutions which are selected to form new solutions (offspring) are selected according to their fitness — the more suitable they are, the more chances they have to reproduce.
- Repeated until some condition is satisfied.

Algorithmic Form

```
Initialize population with random candidates,  
Evaluate all individuals,  
While termination criteria is not met  
    Select parents,  
    Apply crossover,  
    Mutate offspring,  
    Replace current generation,  
end while
```

Termination Criteria

- A specified number of generations or fitness evaluations (100 or 150 generations).
- A minimum threshold reached.

- No improvement in the best individual for a specified number of generations.

Genetic Crossover and Genetic Mutation

The source illustrates genetic crossover as producing offspring that have combinations of features inherited from each parent.

The source illustrates genetic mutation as small random changes being observed.

Image adapted from <http://www.wpdipart.com>

Encoding in Genetic Algorithms (GA)

Encoding in Genetic Algorithms (GA) refers to the way solutions (individuals) are represented in a format that the algorithm can process. Since GAs work with populations of solutions, each solution needs to be encoded as a chromosome (or genotype) before applying selection, crossover, and mutation.

Binary Encoding

The process of representing the solution in the form of a bit string.

```
101101100011
010011001100
```

Permutation Encoding

Every chromosome is a sequence of numbers, which represents the number in the sequence. Used in ordering problems.

Example: Traveling Sales Person Problem. Encoding: chromosome represents the order of cities to visit them.

The source shows example chromosomes including: A: 1153264798 and B: 856723149.

Value Encoding in Genetic Algorithm (GA)

Value encoding is a way to represent solutions in a numeric form instead of binary (0s and 1s). It is useful when the solution involves real numbers, decimal values, or symbols.

What is Value Encoding?

- Each chromosome is a string of real numbers or symbols.
- Used when binary encoding is not suitable (e.g., in complex optimization problems).

Neural Network Weight Optimization

Chromosome stores real-valued weights:

```
[0.25, -0.85, 0.63, 1.23, -0.44]
```

GA Operators

Genetic Algorithms (GAs) use three main operators to evolve better solutions:

- Selection (Choosing Parents)
- Crossover (Combining Solutions)
- Mutation (Introducing Small Changes)

Reproduction / Selection

- Chromosomes are selected from the population as parents to crossover and produce offspring.
- Also known as Selection Operator.
- Parents are selected according to their fitness.
- There are many methods to select chromosomes.
- Roulette Wheel Selection.
- Steady State Selection.
- Elitism.
- The better the chromosomes are, the more chances they have to be selected.

Crossover

Applied to create a better string.

Proceeds in three steps:

- The reproduction operator selects at random two individual strings for mating.
- A cross-site is selected at random along the length.
- The position values are swapped between the parents following the cross-site.

If offspring is not good, it will be removed in the next iteration during "Selection".

If parents are good, the probability of the offspring being good is high.

The source explains that crossover and mutation serve different roles: crossover selects bits for the purpose of "exploitation", whereas mutation serves the purpose of "exploration".

Single Point Crossover

- A cross-site is selected randomly along the length of the mated strings.
- Bits next to the cross-site are exchanged.
- If good strings are not created by crossover, they do not survive beyond the next generation.

Two-point Crossover

- Two random sites are chosen.
- The contents bracketed by these are exchanged between two mated parents.

Uniform Crossover

- Bits are randomly copied from the first or from the other parent.
- A random mask is generated.
- The mask determines which bits are copied from one parent and which from the other parent.

```
Mask:      0110011000
Parent 1:  070001110
Parent 2:  0011010010
```

Mutation

- Restores lost genetic materials.
- Mutation of a bit involves flipping it, changing 0 to 1 and vice versa.

The source contains diagrams showing offspring results for crossover and mutation.